

Week 8: Waste Disposal and Recycling

JSS3 Intermediate Science — First Term Lesson Note (Deep-Dive Edition)

Introduction

This week examines how waste is disposed of, why recycling matters, and the consequences of poor waste management for both the environment and public health.

Methods of Waste Disposal

Method	Description
Landfill	Burying waste in designated land areas
Incineration	Burning waste at high temperatures to reduce its volume
Composting	Allowing organic waste to decompose naturally into fertile soil material
Recycling	Processing used materials into new products
Open dumping (unsafe)	Discarding waste in open, unmanaged areas — not a safe method, but common where proper systems are lacking

Importance of Recycling

- Reduces the amount of waste sent to landfills
- Conserves natural resources by reusing materials
- Saves energy compared to producing new materials from raw resources
- Reduces pollution and litter in the environment
- Can create income opportunities through the sale of recyclable materials

Practical Waste Sorting Activity

As a class activity, sort a sample of household waste into categories: **organic waste** (food scraps, garden waste), **plastic**, **paper/cardboard**, **glass**, and **metal**. Discuss which categories can be recycled or composted, and which require careful disposal.

Effects of Poor Waste Management

- Blocked drainage systems, leading to flooding
- Spread of disease through pests attracted to waste (flies, rodents)
- Water and soil pollution from leaking waste materials
- Air pollution from uncontrolled burning of waste
- Reduced aesthetic and property value of affected areas

Real-Life Application

Blocked gutters filled with plastic waste are a major contributor to urban flooding in many Nigerian cities during the rainy season, illustrating the direct link between waste management and community safety.

Vocabulary

Word	Meaning
Landfill	A site where waste is buried for disposal
Incineration	The controlled burning of waste
Compost	Decomposed organic material used to enrich soil

Evaluation Questions

- 1 List three methods of waste disposal.
- 2 Explain the difference between composting and incineration.
- 3 Give three benefits of recycling.
- 4 Name the five waste categories used in the sorting activity.
- 5 Which type of waste is suitable for composting?
- 6 Describe two effects of poor waste management.
- 7 How can blocked drainage from waste contribute to flooding?
- 8 Why is open dumping considered an unsafe disposal method?
- 9 Give a real-life example of poor waste management you have observed in your community.
- 10 In your own words, explain why recycling benefits both the environment and the economy.